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MILITARY SPECIFICATION

REPEATER, TELEPHONE (REPEATER AN/TCC-11( ))

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of compact, unattended telephone repeater that works over non-loaded spiral-four cable.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-571 — Solder; Soft (Tin, Tin-Lead, and Lead-Silver).

UU-T-111 — Tape; Paper, Gummed (Sealing, and Securing).

LLL-F-291 — Fiberboard; Corrugated, Single-Face (Flexible).

PPP-B-621 — Boxes, Wood Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof, for Packaging and Sealing.

MILITARY

MIL-P-75 — Packaging, Packing, and Container-Marking of Electron Tubes.

MIL-P-116 — Preservation, Methods of.

MIL-P-117 — Bags, Interior Packaging.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures, and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment, and Storage.

MIL-STD-130 — Identification Marking of U. S. Military Property.

FSC 5805

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MIL–STD–169 — Extreme-Temperature Cycle.

MIL–STD–170 — Moisture Resistance Test Cycle for Ground Signal Equipment.

MIL–STD–252 — Wired Equipment, Classification of Visual, and Mechanical Defects for.

DRAWINGS

SIGNAL CORPS

SC–DL–64180 — Repeater, Telephone (Repeater AN/TCC–11( )).

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H–28 — Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification.
Consolidated Freight Classification.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 **Description.** Repeater, Telephone AN/TCC–11( ) is an unattended 12-channel carrier telephone system repeater designed for use with companion equipments Terminal, Telephone AN/TCC–7( ); Repeater, Telephone AN/TCC–8( ) and Test Set. Telephone TS–712( )/TCC–11 over non-loaded spiral-four cable. It is cylindrical in shape and measures 10 inches diameter by 28¼ inches long.

3.2 **Construction.** Repeater, Telephone AN/TCC–11( ) shall be constructed in accordance with the Drawing and Data List SC–DL–64180 and shall meet the requirements of this specification. In case of a conflict between the specification and the drawing, the specification shall govern.

3.3 **Preproduction samples.** The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract (see 6.2).

3.4 **Parts, materials, and processes; general.**

3.4.1 Parts, materials, and processes shall be such as to enable the equipment to meet the performance and other requirements, as specified herein.

3.4.2 *Plastic materials.* Where not machined, plastic material shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the kind of material used and intended application.

3.4.3 *Soldering.*

3.4.3.1 *Solder.* Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ–S–571.

3.4.3.2 *Flux and cleaning agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections, and for tinning or solder-

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ing of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.3.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged, and parts fastened thereto shall not have become loosened.

3.4.4 *Cleaning.* Soldered, and brazed parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document, removing all solder particles, flux and other foreign materials from components. Welded parts and assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing or other suitable means. Cleaning process shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment.

3.5 Electrical requirements.

3.5.1 *Continuity.* Electrical continuity of wiring and other connections, including ground connections, shall conform to the Drawing and Data List SC–DL–64180 and shall be verified by the test of 4.5.1.

3.5.2 *Dc resistance on filter section before the repeater is assembled* (see 4.5.2.1).

Measurement		Dc resistance, ohms		
From Pin	*To Pin*	*Avg.*	*Min.*	*Max.*
8 of J7	11 of J8	5900	5300	6500
8 of J7	13 of J7	1100	1000	1200
13 of J7	20 of J7	1200	1000	1300
20 of J7	11 of J7	1200	1100	1300
11 of J7	13 of J8	0	0	1
13 of J8	20 of J8	1200	1100	1300
20 of J8	11 of J8	1200	1100	1300

3.5.3 *Dc resistance with repeater assembled.*

3.5.3.1 Under the conditions specified in 4.5.2.2 the dc resistance between pin D and pin M on J3 shall be greater than 5 megohms.

3.5.3.2 Under the test conditions described in 4.5.2.3 dc resistance measurements shall be as follows:

Measurement		Dc resistance, ohms		
From terminals *	*To terminals* *	*Avg.*	*Min.*	*Max.*
2 of T1	2 of T4	1200	1000	1400
2 of T3	2 of T2	34	20	50
2 of T1	2 of T2	0.1 Meg.	0.08 Meg.	0.12 Meg.
2 of T3	2 of T4	0.1 Meg.	0.08 Meg.	0.12 Meg.

* T1, T2, T3, and T4 are test circuit repeating coils, figure 1.

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3.5.3.3 Under the test conditions of 4.5.2.4 dc. resistance measurements shall be as follows:

From Terminal	*To Terminal*	*Avg.*	*Min.*	*Max.*
2 of T1	2 of T4	Open ckt.	...	...
2 of T3	2 of T2	Open ckt.	...	...
2 of T1	2 of T2	1200	1000	1400
2 of T3	2 of T4	5.0	3.5	6.5

3.5.4 *Relay operation.* For proper relay operation, a dc-resistance measurement made in accordance with 4.5.3 should indicate a resistance equal to or greater than 1 megohm.

3.5.5 *Current.* The total current drawn from the 200-v power supply shall be 100 ±7 ma. when tested in accordance with 4.5.4.

3.5.6 *Dc voltage requirements.* The dc voltage requirements are as follows for test conditions of 4.5.5:

Measurement		Voltage requirement		
From pin	*To pin*	*Avg.*	*Min.*	*Max.*
J on J3	M on J3	A = 15.8	14.8	16.8
B on J3	M on J3		A + 0.4	A + 2.7
K on J3	M on J3		A + 0.4	A + 2.7
F on J3	M on J3		A + 0.4	A + 2.7
P on J3	M on J3		A + 0.4	A + 2.7

3.5.7 *Transmission.*

3.5.7.1 *Insertion gain of amplifier No. 1 with the gain switch on the AVG position.* The insertion gain requirements, at an output at the OUT 1 jack, figure 1, of 0 dbm, for the ambient temperature existing at time of test, shall be as follows when tested in accordance with 4.5.6.1:

Freq. kc	*Temp. °F.*	*Gain. db*	*Deviation from 68 kc gain. db*
68	50	22.3 ± 0.7	...
68	55	22.4 ± 0.7	...
68	60	22.5 ± 0.7	...
68	65	22.7 ± 0.7	...
68	70	22.8 ± 0.7	...
68	75	22.9 ± 0.7	...
68	80	23.0 ± 0.7	...
68	85	23.1 ± 0.7	...
68	90	23.2 ± 0.7	...
68	95	23.3 ± 0.7	...
68	100	23.5 ± 0.7	...
68	105	23.6 ± 0.7	...
68	110	23.7 ± 0.7	...
28	...	...	— 4.7 ± 0.6
12	...	...	— 6.5 ± 0.5

3.5.7.2 *Insertion gain of Amplifier No. 1 with the gain switch on the high position.* With the GAIN switch of Amplifier No. 1 in the HIGH position the deviation from the 68 kc gain of 3.5.7.1 shall be between +2.2 and +2.6 db (see 4.5.6.2).

3.5.7.3 *Insertion gain of Amplifier No. 1*

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with the GAIN switch on the LOW position. With the GAIN switch of Amplifier No. 1 in the LOW position the deviation from the 68 kc gain of 3.5.7.1 shall be between —2.4 and —2.8 db (see 4.5.6.3).

3.5.7.4 *Insertion loss.* The 1,000 cps insertion loss of the repeater, measured as specified in 4.5.6.4, shall be 2.4 db ±0.5 db.

3.5.7.5 *Insertion gain and insertion loss of Amplifier No. 2.* The requirement for the insertion gain, and insertion loss of Amplifier No. 2 are the same as the requirements for Amplifier No. 1 in 3.5.7.1, 3.5.7.2, 3.5.7.3, and 3.5.7.4 (see 4.5.6.5).

3.5.8 *Looping filter transmission.* Under the test conditions described in 4.5.7 results shall be as follows:

Rep (S2) switch		Output, dbm		
Freq. kc	*In Position*	*Avg.*	*Min.*	*Max.*
68	3	...	...	— 65
83	3	— 1.0	— 6.0	+ 4.0
83	2	...	...	— 43
91	3	...	...	— 43
91	2	+ 2.0	— 3.0	+ 7.0
91	1	...	...	— 34
99	2	...	...	— 34
99	1	+ 4.0	— 1.0	+ 9.0
110	1	...	...	— 34

3.5.9 *Test jack ac-voltages.* The ac-voltage measurements made in accordance with 4.5.8 shall be as follows:

Measurement		Ac voltage
From pin	*To pin*	*Requirements*
C of J3	M of J3	0.85 ± 2 db
A of J3	M of J3	0.12 ± 2 db

3.5.9.1 The ac-voltage measurements made in accordance with 4.5.8.1 shall be as follows:

Measurement		Ac voltage
From pin	*To pin*	*Requirements*
L of J3	M of J3	0.55 V. ± 2 db

3.5.9.2 The ac-voltage measurements made in accordance with 4.5.8.2 shall be as follows:

Measurement			Ac voltage
Freq. kc	*From pin*	*To pin*	*Requirements*
12	E of J3	M of J3	0.85 v. ± 2 db
12	R of J3	M of J3	0.12 v. ± 2 db
1	N of J3	M of J3	0.55 v. ± 2 db

3.5.10 *Crosstalk.* Under the test conditions specified in 4.5.9, 4.5.9.1, 4.5.9.2 and 4.5.9.3 measurements shall meet the following requirements:

Freq. kc	*Loss requirement between equal level points, db*
12	Equal to or greater than 68
28	Equal to or greater than 68
68	Equal to or greater than 68

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3.5.11 *Precision gain requirements.* The manually equalized transmission characteristics of a repeater section (5.75 miles of CX–1065/G cable and one Telephone Repeater AN/TCC–11( )) obtained in accordance with 4.6 shall not deviate more than ±0.05 db from the transmission characteristics in figure 6.

3.5.12 *System use.* The equipment shall be capable of operating in a system with Telephone Terminal AN/TCC–7( ) and with Telephone Repeater AN/TCC–8( ) (see 4.7).

3.6 **Service conditions.** The equipment shall meet the following service conditions:

3.6.1 *Equipment nonoperating.* The equipment shall comply with the operational requirements of 3.6.2 after subjection to any of the following nonoperating conditions successively, or in combinations encountered during worldwide short-term storage and transit:

(a) *Temperature.* Continuous exposure for 24 hours at ±160°F. and 24 hours at —80°F. (see 4.8.1).

(b) *Elevation.* Elevation up to 25,000 feet above sea level (see 4.8.2).

(c) *Relative humidity.* Relative humidity of 100%, including condensation caused by temperature changes (see 4.8.3).

(d) *Immersion* (see 4.8.4).

(e) *Vibration and shock* (see 3.6.3).

3.6.2 *Equipment operating.* The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) *Temperature.* Any ambient temperature in the range of —40°F. to +125°F.

(b) *Elevation.* Any elevation up to 10,000 feet above sea level.

(c) *Relative humidity.* Any relative humidity up to 95 percent.

3.6.3 *Vibration and shock.*

3.6.3.1 *Vibration.* Except for internal resonance of parts and subassemblies specified herein, the equipment shall have no mechanical resonance below 55 cycles per second, and shall conform to the requirements of 4.8.5.

3.6.3.2 *Shock; drop.* The equipment shall be capable of withstanding the shock test of 4.8.6 without evidence of degradation of specification performance.

3.7 **Preconditioning.** The equipment shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.8 **Marking.**

3.8.1 *General.* Marking shall conform to Standard MIL–STD–130.

3.8.2 *Serial numbers.* The equipment shall be serial numbered.

3.9 **Interchangeability.** Corresponding units and replaceable assemblies, subassemblies, and parts shall be physically and functionally interchangeable as complete items without modification thereof, or of other articles with which the items are used (see 4.9). When dimensions, ratings, chararteristics, etc., are not specified the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly, or part is to be considered replaceable, the contracting officer shall be consulted.

3.10 **Workmanship.** Workmanship shall be such as to meet all applicable requirements of this specification and any referenced subsidiary specification, or other document when

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inspected in accordance with section 4 and the following paragraphs:

3.2 **Construction.**

3.4.2 *General requirements for plastic material and plastic parts.*

3.4.3 *Soldering.*

3.4.4 *Cleaning.*

3.8 **Marking.**

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility. Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 *Classification of inspection.* Inspection shall be classified as follows:

(a) *Preproduction inspection.* Does not include preparation for delivery (see 4.2).

(b) *Procurement inspection.* Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.11.

(1) Procurement inspection of items before preparation for delivery (see 4.3).

(2) Procurement inspection of preparation for delivery (see 4.11 and section 5).

4.2 Preproduction inspection. This inspection will be performed by the Government unless otherwise specified in tables I, II, and III. It shall consist of the nondestructive group A, group B, and group C examinations and tests specified in tables I, II, and III respectively. The preproduction inspection will normally be performed in this order: (1) vibration; (2) shock, drop, and (3) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

4.3 Procurement inspection of items before preparation for delivery. The contractor shall perform the inspection specified by (a) through (d) below, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government. at its discretion — may perform all or any part of the inspection specified in (a) to verify the contractor's compliance with specfied requirements. The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection (see 6.7).

(a) Procurement inspection shall consist of group A, group B, and group C inspection as specified in 4.3.1 through 4.3.3.2.

(b) When Standard MIL–STD–105 specifies actions by Government, the Government may authorize the contractor to perform any of such actions except that the responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection, and on samples selected from units that have been sub-

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jected to, and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to, and met group B inspection. However, the order may be varied when it is considered more practical to select separate samples for group B, or group C inspection, or both.

(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly (see 3.7).

4.3.1 *Group A inspection.* This inspection (including sampling) shall conform to table I and Standard MIL–STD–105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the system inspection (4.7) shall be last.

TABLE I. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL		
			Major	Minor	Control
Visual and mechanical	3.10	4.10	10 dphu	25 dphu	100 dphu
Electrical tests					
Continuity	3.5.1	4.5.1			
Dc Resistance	3.5.2	4.5.2.1			
Dc Resistance	3.5.3.1	4.5.2.2			
Dc Resistance	3.5.3.2	4.5.2.3			
Dc Resistance	3.5.3.3	4.5.2.4			
Relay operation	3.5.4	4.5.3			
Current	3.5.5	4.5.4			
Dc Voltage	3.5.6	4.5.5		*	
Insertion gain of Amplifier No. 1	3.5.7.1	4.5.6.1	15 dphu for the group		
	3.5.7.2	4.5.6.2			
	3.5.7.3	4.5.6.3			
Insertion loss of Amplifier No. 1	3.5.7.4	4.5.6.4			
Insertion gain and loss of Amplifier No. 2.	3.5.7.5	4.5.6.5			
Looping filter transmission	3.5.8	4.5.7			
Test jack ac voltages	3.5.9	4.5.8			
Ac voltage measurements	3.5.9.1	4.5.8.1			
	3.5.9.2	4.5.8.2			
Crosstalk	3.5.10	4.5.9			
		4.5.9.1			
		4.5.9.2			
		4.5.9.3			

* All electrical tests are considered major.

4.3.2 *Group B inspection.* This inspection (including sampling) shall conform to table II and the appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. Disposition of the nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.5 and the requirements of Standard MIL–STD–105 for disposition of rejected product.

4.3.2.1 *Group B sampling plans.* The group B sampling plans, as listed in table II, shall be as follows:

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Group B plan	*AQL*	*Inspection level for normal inspection*	*Inspection level for reduced inspection*
B–2	6.5%	L–7	L–5

4.3.2.2 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE II. *Group B inspection*

Inspection	Requirement paragraph	Inspection paragraph	Group B plans (see 4.3.2.1)
Interchangeability	3.9	4.9	B–2
System test	3.5.12	4.7	B–2

4.3.3 *Group C inspection.* This inspection shall be as listed in table III.

4.3.3.1 *Sampling for inspection of equipment.* One each AN/TCC–11( ) for each group C inspection shall be selected out of every fifty submitted for acceptance, but not more than two samples each per month, without regard to quality.

4.3.3.2 *Noncompliance.* If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process, and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.4).

TABLE III. *Group C inspection*

Inspection	Requirement paragraph	Inspection paragraph
Altitude	3.6.1 and 3.6.2	4.8.2
Humidity	3.6.1 and 3.6.2	4.8.3
Immersion	3.6.1	4.8.4
Shock test: drop	3.6.3.2	4.8.6
Temperature	3.6.1 and 3.6.2	4.8.1
Vibration	3.6.3.1	4.8.5
Precision gain	3.5.11	4.6

4.3.4 *Reinspection of conforming group B and group C sample units.* Unless otherwise specified, sample units which have been subjected to, and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.5 *Disposition of nonconforming product.* The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and the means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A or B inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted.

4.4 Bounce preconditioning. The unit shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, N. Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be sub-

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jected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.5 Electric tests. The equipment shall be inspected and tested to determine compliance with the requirements of 3.5 through 3.5.12 as follows:

4.5.1 *Continuity.* Conventional ohm meters may be used for checking continuity in this equipment provided the short-circuit current between the test leads of the ohm meter dos not exceed 200 milliamperes and the product of the open-circuit voltage and the short-circuit current does not exceed 0.4 volt-ampere on any scale (see 3.5.1).

4.5.1.1 Continuity tests shall be made on all wiring except where lack of continuity, or incorrect association of apparatus will cause a failure of some other test. Sufficient continuity tests shall be made in all cases to insure that the electrical association of apparatus and equipment is in accord with the equipment layout and with the apparatus and equipment designations.

4.5.1.2 A check shall be made to insure that paired and shielded wire is used where specified and that the shielding is correctly connected to the terminals specified.

4.5.1.3 All terminals and all apparatus cases for which ground is specified shall be checked to insure that ground is present.

4.5.1.4 All wires that are enclosed in shielding and all apparatus cases that are insulated from the panel shall be tested for the absence of ground.

4.5.1.5 The proper functioning of all keys and controls shall be assured primarily through transmission tests and not by visual inspection alone.

4.5.2 *Dc resistance.*

4.5.2.1 A dc resistance test shall be made on the filter section before the repeater is assembled to assure that the correct voltages will be applied to the amplifiers. The dc-resistance measurements shall meet the requirements of 3.5.2.

4.5.2.2 With the repeater assembled a dc-resistance measurement between pin D and pin M on J3 shall meet the requirements of 3.5.3.1. This test shall be made with the power switch in both PWR LOOP and PWR THRU position.

4.5.2.3 With the repeater connected to the test circuit shown in figure 1, but with connections to the 200-v power supply and coil LI omitted and the power switch in the PWR THRU position, the dc-resistance measurements shall meet the requirements of 3.5.3.2.

4.5.2.4 With connections as in 4.5.2.3, but with the power switch in the PWR LOOP position, the dc-resistance measurements shall meet the requirements of 3.5.3.3.

4.5.3 *Relay operation.* With the repeater connected to the test circuit, but with the regular 200-v power connections omitted and with the power switch in the PWR THRU position, relay operating current should be applied between terminal 2 of T3 and terminal 2 of T2, as shown in figure 1. For proper rleay operation, a dc-resistance measurement between pin 2 of T1 and pin 2 of T2 should indicate a resistance equal to or greater than 1 megohm as required by 3.5.4.

4.5.4 *Current.* With the repeater connected to the test circuit as shown in figure 1, set the power switch in the PWR LOOP position and apply power. The total current drawn shall be within the limits of 3.5.5.

4.5.5 *Dc voltage.* With the repeater connected to the test circuit shown in figure 1, dc-voltage measurements shall be made to determine compliance with 3.5.6.

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4.5.6 *Transmission.*

4.5.6.1 *Insertion gain of Amplifier No. 1 with the GAIN switch on the AVG position.* The insertion gain of the repeater shall be measured on a 135 ohm circuit as shown in figures 1 and 2. The 600 ohm terminations shall be removed from the IN I and OUT 1 jacks of the test circuit. The insertion gain measurements at an output at the OUT I jack, figure 1, of 0 dbm for the ambient temperature existing at the time of test shall meet the requirements of 3.5.7.1.

4.5.6.2 *Insertion gain of Amplifier No. 1 with the GAIN switch on the HIGH position.* Measurements shall be made to determine the deviation from the 68 kc gain of 4.5.6.1. The deviation shall be within the limits of 3.5.7.2.

4.5.6.3 *Insertion gain of Amplifier No. 1 with the GAIN switch in the LOW position.* Measurements shall be made to determine the deviation from the 68 kc gain of 4.5.6.1. The deviation shall be within the limits of 3.5.7.3.

4.5.6.4 *Insertion loss of Amplifier No. 1.* The 1,000 cps insertion loss shall be measured in the 135 ohm circuit shown in figures 1 and 2. The measurement shall meet the requirements of 3.5.7.4.

4.5.6.5 *Insertion gain and insertion loss of Amplifier No. 2.* The tests of 4.5.6.1, 4.5.6.2, 4.5.6.3, and 4.5.6.4 shall be repeated between the IN 2 and OUT 2 jacks of the test circuit in figures 1 and 2. GAIN switch settings of Amplifier No. 2 shall be varied as for Amplifier No. 1. Measurements shall meet the requirements of 3.5.7.5.

4.5.7 *Looping filter transmission.* With the P2 connector, of the test circuit in figure 1, removed from J2 of the repeater, 135 ohm terminations shall be connected to the input and output circuits of J2 of the repeater. With —25 dbm signals applied to the IN 1 jack the output at the OUT 2 jack shall meet the requirements of 3.5.8.

4.5.8 *Test jack ac voltage.* With a signal of 12 kc applied to the IN I jack and the level adjusted so that the output at OUT I is 0 dbm, the ac voltage measurements shall meet the requirements of 3.5.9.

4.5.8.1 With conditions as in 4.5.8 but at a frequency of 1 kc at —6 dbm at OUTPUT I, ac-voltage measurements shall meet the requirements of 3.5.9.1.

4.5.8.2 With test conditions as described in 4.5.8 and 4.5.8.1, but with the test signals applied at the IN 2 jack, figure 1, ac-voltage measurements shall meet the requirements of 3.5.9.2.

4.5.9 *Crosstalk.* The dc-power leads shall be removed from the test circuit (fig. 1) at T1 and T2. Power shall be applied to the repeater as shown in figure 3. P1 of the test circuit shall be connected to J1 of the repeater, and P2 of the test circuit to J2 of the repeater. The REP (S2) switch shall be in position 3 and the power switch in the PWR THRU position. The signal into IN I shall be adjusted so that the output at OUT I is 0 dbm into 600 ohms. The test circuit connector P2 shall be removed from J2 of the repeater and the repeater input and output circuits at J2 shall be terminated in 135 ohms. With the OUT 2 jack terminated in 600 ohms the output at the OUT 2 jack shall meet the requirements of 3.5.10.

4.5.9.1 The measurements of 4.5.9 shall be repeated with the power switch in the PWR LOOP position and shall meet the requirements of 3.5.10.

4.5.9.2 The power switch shall be set in the PWR THRU position. The terminations shall be removed from the J2 connector of

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the repeater and P2 of the test circuit (fig. 1) shall be connected to J2 of the repeater. P1 of the test circuit shall be removed from J1 and 135 ohm terminations shall be connected to the input and output circuits of J1. With the OUT 2 jack terminated in 600 ohms, the output at the OUT 2 jack shall meet the requirements of 3.5.10.

4.5.9.3 The measurements of 4.5.9.2 shall be repeated with the power switch in the PWR LOOP position and shall meet the requirements of 3.5.10.

4.6 Precision gain measurements. The procedure for checking a repeater for the precision gain requirements of 3.5.11 is as follows:

4.6.1 Place the repeater in a temperature chamber, apply power to the repeater, set the temperature to +45°F. and allow the repeater to soak for at least 6 hours.

4.6.2 Measure the insertion gain of the repeater between CX–1065/G cable impedance (135 ohms) to an accuracy of at least ±0.005 db over the frequency range from 12 kc to 68 kc inclusive.

4.6.3 Calculate the transmission frequency characteristics of a repeater section by subtracting the +45°F. loss of 5.75 miles of CX–1065/G cable (fig. 4), from the gain measurements of 4.6.2.

4.6.4 Apply manual equalization to the transmission frequency characteristic following the procedure given in figure 5.

4.6.5 Compare the curve obtained by 4.6.4 to the reference curve in figure 6. The maximum deviation from the reference curve shall meet the requirements of 3.5.11.

4.7 Operations test. The AN/TCC–11( ) repeater shall be tested for satisfactory operation by connecting it in a telephone carrier system between two AN/TCC–7 terminals (Terminal A and Terminal B), using the equivalent of 5¾ miles of spiral four cable between each AN/TCC–7 terminal and the AN/TCC–11( ) repeater. Telephone conversation between terminals shall have satisfactory volume and intelligibility in both the A to B and the B to A direction.

4.8 Service conditions test. The following service conditions test will be performed on the equipments.

4.8.1 *Temperature test.* The equipment shall be subjected to the temperature cycle shown on Standard MIL–STD–169 and shall meet the requirements of 3.5 through 3.5.11 at steps 1, 4, 8, and 10.

4.8.2 *Altitude test.* The equipment shall be placed in an altitude chamber at ambient conditions. Performance shall then be measured for compliance with paragraph 3.5.7. The pressure shall then be reduced equal to 10.7 inches of mercury and stabilized for a period of two hours. The pressure shall then be raised equal to 21 inches of mercury and the chamber maintained at this pressure for another period of two hours. The performance shall again be measured at this altitude. The pressure shall then be raised to ambient conditions and the performance again measured. There shall be no degradation of performance in the second and third measurements from those of the first measurement (see 3.6.1 and 3.6.2).

4.8.3 *Moisture-resistance test for equipment.* The equipment shall be tested as follows (see 3.6.1 and 3.6.2).

4.8.3.1 *Test conditions.*

(a) Do not remove equipment from humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete test measurements as rapidly as possible. Disconnect from power

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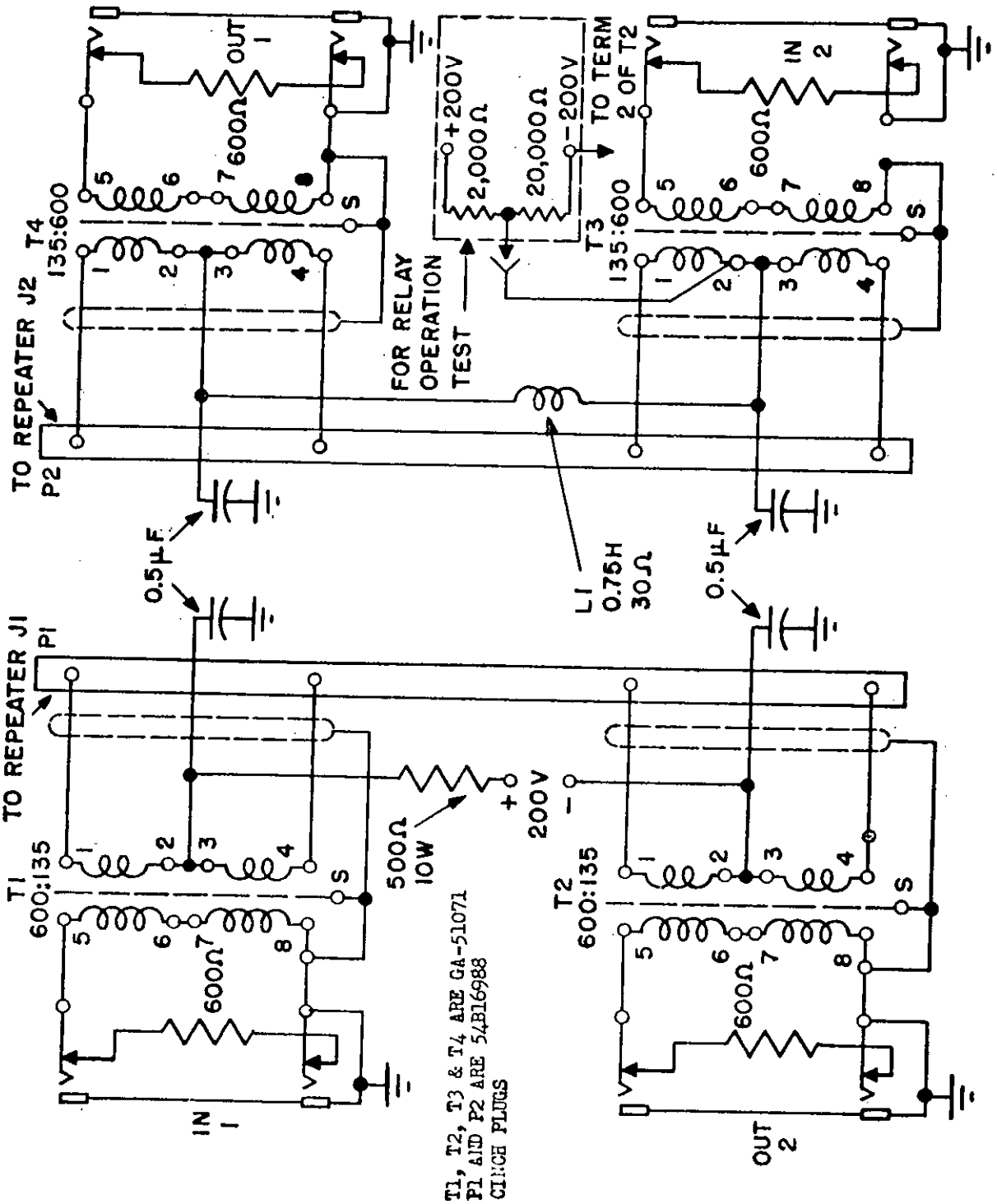


FIGURE 1. *AN/TCC-11 repeater test circuit.*

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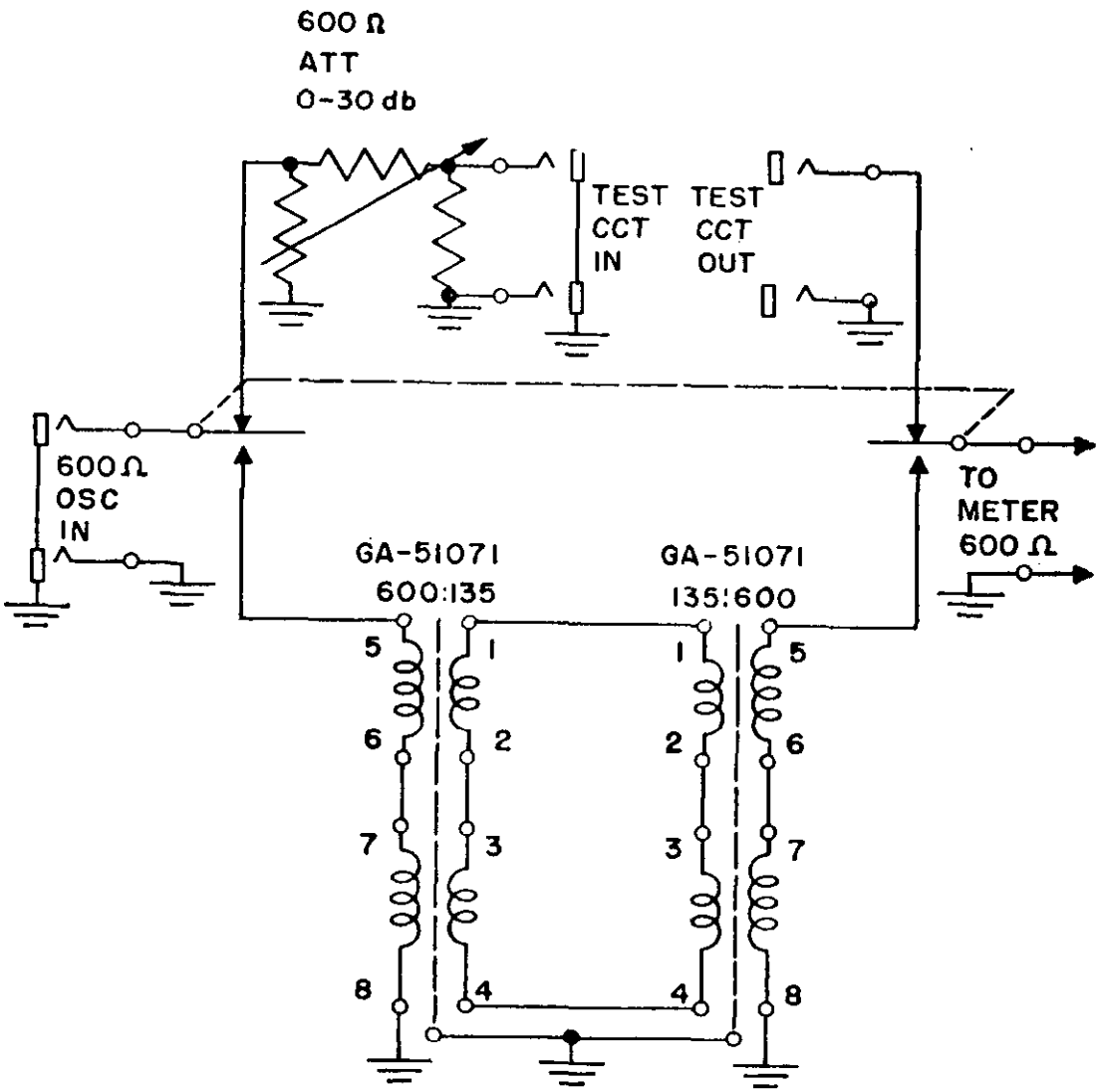


FIGURE 2. *Circuit for measuring insertion gain of AN/TCC-11 repeater.*

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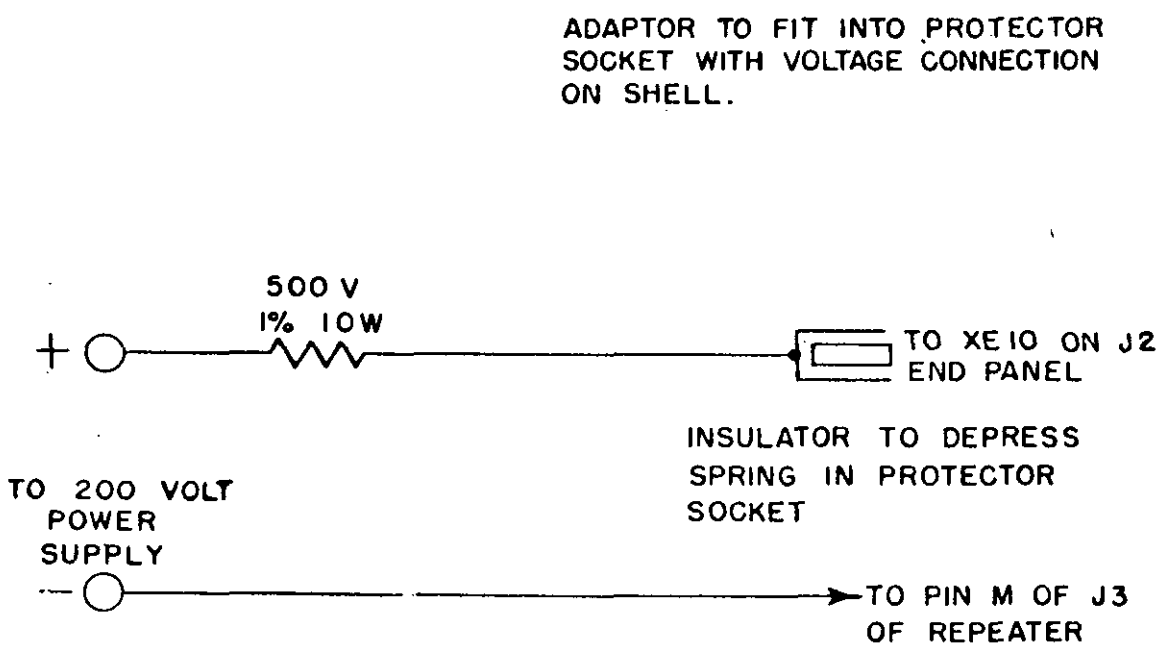


FIGURE 3. *Power connections for crosstalk tests.*

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CX-1065/G CABLE LOSS 5.75 MILES AT 45°F.

FREQ KC	LOSS db	FREQ KC	LOSS db	FREQ KC	LOSS db
		32	17.788	56	20.561
		33	17.866	57	20.697
10	15.240	34	17.984	58	20.834
11	15.466	35	18.084	59	20.973
12	15.663	36	18.186	60	21.113
13	15.838	37	18.288	61	21.254
14	15.988	38	18.392	62	21.395
15	16.126	39	18.498	63	21.538
16	16.254	40	18.607	64	21.681
17	16.361	41	18.718	65	21.829
18	16.475	42	18.829	66	21.981
19	16.584	43	18.944	67	22.133
20	16.680	44	19.059	68	22.286
21	16.776	45	19.177	69	22.438
22	16.869	46	19.295	70	22.591
23	16.961	47	19.414	71	22.742
24	17.050	48	19.535	72	22.897
25	17.142	49	19.658	76	23.512
26	17.231	50	19.782	80	24.204
27	17.321	51	19.909	100	27.002
28	17.411	52	20.037		
29	17.504	53	20.166		
30	17.598	54	20.296		
31	17.692	55	20.428		

FIGURE 4.

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MANUAL EQUALIZATION OF REPEATER SECTION CHARACTERISTIC

A. To Apply Flat Equalization

1. Find the difference between 0 db excess gain and the actual excess gain of the repeater in question at 68 KC. Apply a flat gain vs. frequency characteristic to the excess gain characteristics of the repeater such that the resultant 68 KC excess gain is 0 db.

B. To Apply Slope Equalization

(This can be done graphically or mathematically.)

Graphical Method

1. On the curve obtained in A draw a straight line between the 68 KC gain point and the 12 KC gain point. The difference between the straight line and the excess gain curve is the slope equalized curve.

Mathematical Method

1. Calculate the straight line equation (y=mx+b) which goes thru the 12 KC and 68 KC gain points on the curve obtained in Section A. From this equation, values of the straight line can be calculated for other frequencies. Subtract the straight line values from the excess gain curve values for the desired frequencies; the result is the slope equalized curve.

C. To Apply Bulge Equalization

1. Find the difference between 0 db excess gain and the actual excess gain at 28 KC on the curve obtained in B. Use figure 6 attached to calculate the bulge correction curve. Subtract this bulge correction curve from the excess gain characteristic to obtain the bulge equalized characteristic.

FIGURE 5.

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EQUALIZER (BULGE)

CONVERSIONS

TO OBTAIN BULGE CORRECTION CURVE MULTIPLY 28 KC CORRECTION REQUIRED BY EACH FACTOR BELOW TO OBTAIN RESULTING BULGE CORRECTION AT THAT FREQUENCY.

FREQ KC	FACTOR
12	0.00
14	0.22
16	0.441
20	0.735
24	0.955
28	1.0
32	0.985
36	0.882
40	0.764
44	0.617
48	0.564
52	0.368
56	0.25
60	0.162
64	0.0882
68	0.00

FIGURE 6. *(2 Sheets) Sheet No. 1.*

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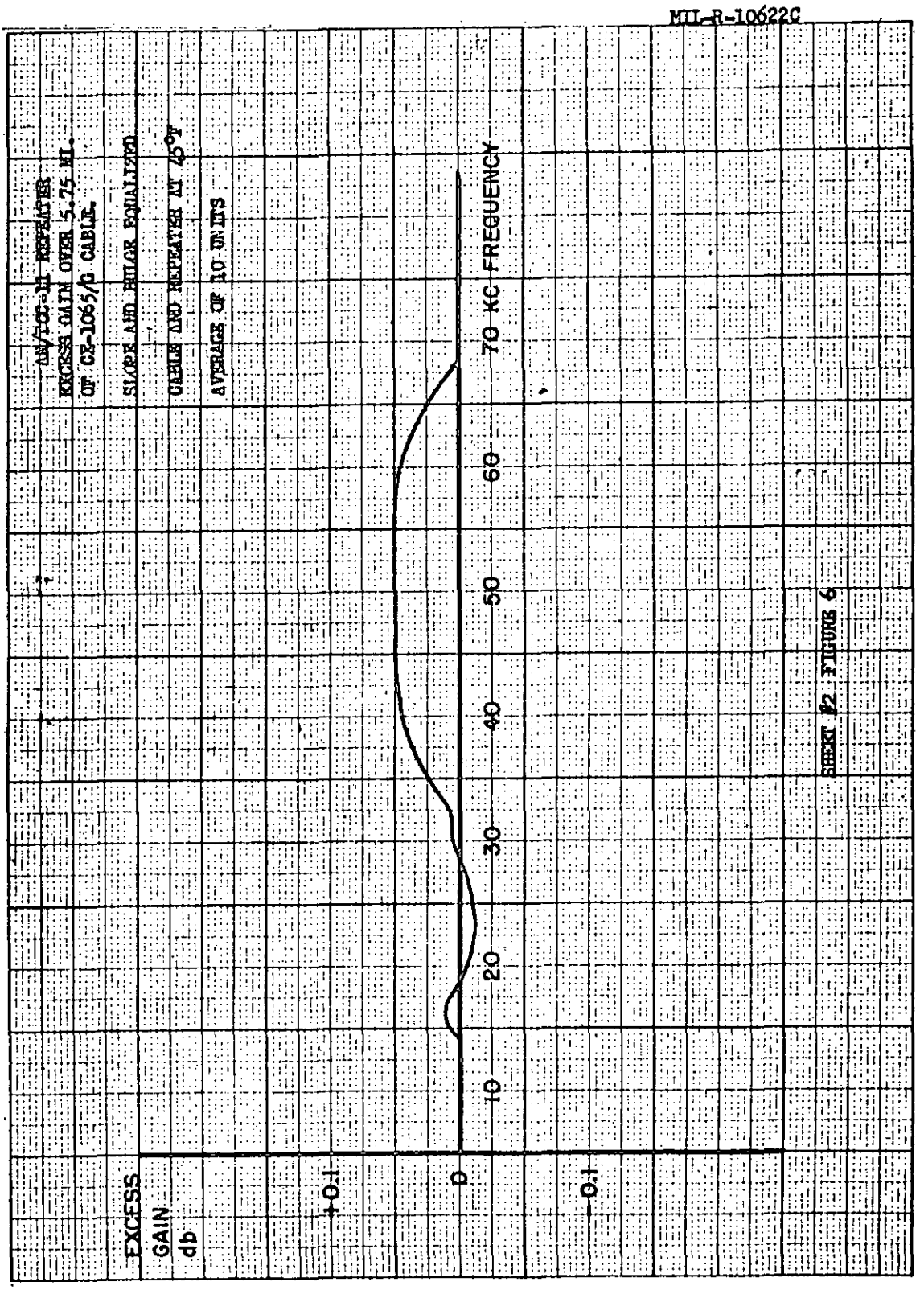


FIGURE 6. *Sheet No. 2.*

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source when measurements are complete.

4.8.3.2 *Test method.*

(a) Dry at 130° ±5°F. for 24 hours.

(b) Condition at 77° ±5°F. and 40-to 50-percent relative humidity for 24 hours.

(c) Test in accordance with 4.5.

(d) Subject to continuous cycling for a total of five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL–STD–170. Repeat test of 4.5.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F. at 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet the full specification requirements of performance. At the completion of the test, the equipment shall be operable and meet the tests of 4.5.

4.8.3.3 *Performance.* The performance and any measurements taken during test shall be equal to the performance before the humidity cycles.

4.8.4 *Immersion test.* The equipment shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion the temperature of the equipment shall be approximately 40°F. above the temperature of the water. The tank in which the sample is immersed shall be of sufficient capacity to maintain the water at approximately its initial temperature, or the temperature of the water shall be maintained constant by other means. After completion of the 2-hour period of immersion, the sample shall be removed from the water and wiped dry on exterior surfaces. There shall be no evidence of leakage (see 3.6.1).

4.8.5 *Vibration test.* The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.6.3.1 as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration (see 4.8.5(d)).

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at $\frac{1}{64}$ inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Mass., or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

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4.8.6 *Shock test; drop.* The equipment shall be dropped a distance of 48 inches for a total of 14 drops. There shall be one drop on each end; 4 drops 90° apart on each edge and 4 drops 90° apart on the side. Drops shall be made from a quick-release hook; or pendulum tester as made by the L.A.B. Corporation, Skaneateles, N. Y., or equal. The floor or barrier receiving the impact shall be 2-inch fir backed by concrete or a rigid steel frame. The equipment shall be operable with no degradation in performance at the end of the test (see 3.6.3.3).

4.9 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive matings parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.10 Visual and mechanical inspection. A thorough inspection shall be made on each completed equipment for mechanical fit, loose nuts, and screws, welding, riveting, finish, poor workmanship and other defects. Fastening devices shall be checked for proper fit and mechanical operation. Welds, finishes, ground connections, and the like shall be checked for satisfactory workmanship and compliance with Standard MIL–STD–252.

4.11 Procurement inspection of preparation for delivery. The contractor shall perform the inspection specified by 4.11.1 and 4.11.2, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the inspection specified in 4.11.1 and 4.11.2, to verify the contractor's compliance with specified requiremnts. The amount of such verification inspection will depend on the extent to which the contractor's inspecion procedures and records insure that the equipment on contract meets the specified procurement inspection (see 6.7). When Standard MIL–STD–105 specifies action by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.11.1 *Presrvation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL–P–116. Classification of defects shall be as shown in table IV.

TABLE IV. *Preservation, packaging, and marking thereof*

Major defects

1. Use of improper or defective material.
2. Quantity in unit package not as specified.
3. Incorrect packaging method applied.
4. Cushioning or padding omitted.
5. Cushioning inadequate for the physical and mechanical protection of the item.
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.
7. Stock number omitted, incorrect, or illegible.

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8. Nomenclature omitted, incorrect, or illegible.
9. Marking of quantity of items in package omitted, incorrect, or illegible.
10. Different stock numbered items in the same unit package.

Minor defects

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Packaging material damaged.
3. Air not expelled from barrier.
4. Insufficient area of barrier material provided to permit 2 subsequent seals.
5. Any item of marking information other than 7 through 10 omitted, incorrect or illegible.

4.11.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by th contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL–STD–105. Inspection level L–8 shall be used for normal inspection and L–6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table V.

TABLE *V. Packing, and marking thereof*

Major defects

1. Use of improper or defective material.
2. Quantity in pack not as specified.
3. Gross weight in excess of specified amount.
4. Box closure not as specified.
5. Type, grade, class, and style of the shipping container not as specified.
6. Strapping omitted.
7. Strapping inadequate or incorrectly applied.
8. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
9. Shipping documents or packing list omitted.
10. Stock number omitted, incorrect, or illegible.
11. Nomenclature omitted, incorrect, or illegible.
12. Marking of quantity of items in pack omitted, incorrect, or illegible.
13. Destination marking omitted, incorrect or illegible.
14. Service designation (color marking) omitted.
15. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
16. Overseas code marking omitted, incorrect, or illegible.

Minor defects

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL–STD–105.
4. Any item of required marking information other than 10 through 16 listed under major defects omitted, incorrect, or illegible.

4.11.2.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.11.2.2 *Disposition of nonconforming product.* Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL–STD–105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure, and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Repeater, Telephone AN/TCC–11( ) shall be presrved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and described in the following paragraphs:

5.1.1.1 *Technical manual.* Each technical manual shall be packaged Method IC–3, as follows: inclose each technical manual within a close-fitting bag, type II, class b, conforming to Specification MIL–B–117. Heat seal to effect closure.

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5.1.1.2 *Tubes (spare).* Spare tubes shall be packaged individually in accordance with the applicable provisions of Specification MIL–P–75.

5.1.1.3 *Arrestors.* Arrestors shall be packaged method III, as follows: items of like description and bearing the same stock number shall be placed together within a wrap of flexible, single-face, corrugated fiberboard, type I, conforming to Specification LLL–F–291. Secure the wrap with gummed paper type, class 1, conforming to Specification UU–T–111.

5.1.1.4 *Stowage within transit case.* Place and secure Repeater, Telephone AN/TCC–11( ) within designated position within its transit case. Stow the items packaged as specified in 5.1.1.2. and 5.1.1.3 in space provided within case. Close lid of case and secure all fastenings.

5.1.1.5 *Transit case with contents.* Each case with contents shall be packaged method IC–5, as follows: cushion each case on all surfaces with cells or pads, or both, fabricated of double-faced, corrugated fiberboard, having a minimum dry bursting strength of 275 pounds, designed to absorb the shock of impact encountered in handling and transit. The cells or pads shall be equal to the water-resistant requirements of th fiberboard box in which they are used. Place the cushioned case within a close-fitting fiberbaord box, style RSC, W5c, conforming to the requirements of Specification PPP–B–636. Closure shall be in accordance with the appendix of the box specification. In addition, seal all seams and joints with waterproof, pressure-sensitive tape, type III, class 1, olive drab, conforming to Specification PPP–T–60.

5.1.2 *Level C.* Repeater, Telephone AN/TCC–11( ) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration, and physical damage, during direct shipment to the first receiving activity.

5.2 Packing

5.2.1 *Level A.* Repeater, Telephone AN/TCC–11( ) shall be packed within a style 4, class 2, nailed, wood box, conforming to Specification PPP–B–621. Fabricate the box to fit contents snugly. Place the technical manual, packaged as specified in 5.1.1.1 between the contents, and lid of the shipping container. Closure shall be in accordance with the box specification.

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped with flat, steel strapping, in conformance with the requirements of the appendix of the container specification, only for direct shipment to ports.

5.2.2 *Level B.* Repeater, Telephone AN/TCC–11( ) shall be packed as specified in 5.2.1 except that the style 4, nailed, wood box shall be class 1, conforming to Specification PPP–B–621. Fabricate the box to fit contents snugly. Closure shall be in accordance with the box specification. Strapping is not required.

5.2.3 *Level C.* Repeater, Telephone AN/TCC–11 shall be packed for shipment in a manner conforming to the requirements of the Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** Repeater, Telephone (Repeater AN/TCC–11( )) is an unattended telephone carrier repeater used in the AN/TCC–7( ) Telephone Carrier Systems. It is designed for field use between telephone terminal (AN/TCC–7( )) stations or be-

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tween attended telephone repeater (AN/TCC–8( )) stations. Repeater, Telephone (Repeater AN/TCC–11( )) stations are placed at 5¾ mile intervals along the spiral-four cable.

6.2 **Ordering data.** Procurement documents *should specify the following:*

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample (see 3.3).

(2) Preproduction pack(s) as follows: detailed instructions for preparation for delivery.

Note. These may be obtained from the Signal Corps Packaging Activity, U. S. Army Signal Supply Agency, Department of the Army, 225 South Eighteenth Street, Philadelphia 3, Pa.

(f) Marking and shipping of samples.

(g) Technical literature required.

(h) Quantity of tools; and running spare parts required.

(i) Government furnished equipment, two complete AN/TCC–7 terminals shall be furnished for the operational test of 4.7.

6.3 **Nomenclature.** The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: AN/TCC–11W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.4 **Group C inspection.** Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of the corrective action (see 4.3.3.2).

6.5 **Definition of level B.** Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, physical damage during domestic shipments, covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.6 **Verification inspection.** The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) *Type A.* The total of that inspection set forth in the Quality Assurance Provisions of this specification, or the contract. Included in this category is that amount of inspection referred to as normal, and tightened inspection by Standard MIL–STD–105.

(b) *Type B.* That inspection set forth in the Quality Assurance Provisions of this specification, or contract reduced in amount under the reduced inspection provisions of Standard MIL–STD–105.

(c) *Type C.* A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality As-

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surance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

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